

Arithmetic Foundation: Fractions Operations & Common Denominators

Exam (Set 7)

Name: _____

Class: _____

Student ID: _____

Time: 35 Mins

Total: 100 Pts

Score: _____

Section 1: Equivalent Fractions & Simplifying (3 pts each, 18 pts total)

Score: _____

Fill in the missing numerators or denominators to make the fractions equivalent, or simplify to lowest terms:

(a) $\frac{3}{7} = \frac{\underline{\hspace{1cm}}}{28}$ _____

(b) $\frac{5}{8} = \frac{35}{\underline{\hspace{1cm}}}$ _____

(c) $\frac{18}{24} = \frac{\underline{\hspace{1cm}}}{4}$ _____

(d) Simplify $\frac{45}{60} =$ _____

(e) Simplify $\frac{28}{42} =$ _____

(f) $\frac{4}{9} = \frac{\underline{\hspace{1cm}}}{36}$ _____

Section 2: Improper Fractions & Mixed Numbers Conversion (3 pts each, 18 pts total)

Score: _____

Express as mixed numbers in simplest form (Quotient $\frac{\text{Remainder}}{\text{Divisor}}$) or convert to improper fractions:

(a) $52 \div 24 =$ _____

(b) $\frac{38}{6} =$ _____

(c) $\frac{47}{5} =$ _____

(d) Convert $4\frac{3}{7}$ to improper fraction: _____

(e) Convert $5\frac{2}{9}$ to improper fraction: _____

(f) $75 \div 12 =$ _____

Section 3: Fraction Addition & Subtraction (Common Denominator) (4 pts each, 24 pts total)

Score: _____

Show your common denominator calculation. Do NOT simply add across numerators and denominators!

(a) $\frac{3}{4} + \frac{2}{5} =$
LCD = 20. Expand BOTH numerators! _____

(b) $\frac{5}{6} - \frac{3}{8} =$
LCD = 24. _____

(c) $2\frac{1}{3} + 1\frac{3}{5} =$
Add whole numbers and fractions. _____

(d) $3\frac{1}{4} - 1\frac{3}{4} =$
Borrow 1 from 3 first! _____

(e) $4\frac{2}{7} - 2\frac{5}{7} =$
 $4\frac{2}{7} = 3\frac{9}{7}$ _____

(f) $\frac{7}{10} - \frac{1}{4} + \frac{2}{5} =$
Find LCD of 10, 4, 5 (LCD = 20). _____

Section 4: Fraction Multiplication & Division (4 pts each, 20 pts total)**Score: _____**

Simplify before multiplying where possible. For division, multiply by the reciprocal (Keep-Change-Flip):

(a) $\frac{4}{9} \times \frac{3}{8} =$ _____

(b) $\frac{5}{12} \times 16 =$ _____

(c) $\frac{7}{10} \div \frac{14}{15} =$ _____

(d) $6 \div \frac{3}{4} =$ _____

(e) $2\frac{1}{4} \times 1\frac{1}{3} \div \frac{9}{10} =$ (Convert to improper fractions first!) _____

Section 5: Comparing Fractions & Word Applications (5 pts each, 20 pts total)**Score: _____**

1. [Comparing Fractions] Insert $>$, $<$, or $=$ in each blank:

(a) $\frac{5}{12}$ _____ $\frac{7}{16}$

(b) $\frac{11}{12}$ _____ $\frac{12}{13}$ (Residual method)

(c) $\frac{7}{9}$ _____ $\frac{7}{11}$ (Same numerator)

2. [School Practical Problem - Recipe Partition]

A baking recipe requires $2\frac{1}{4}$ kg of flour. Student A has a bag containing $\frac{3}{5}$ kg of flour and Student B brings another bag containing $1\frac{1}{2}$ kg of flour.

(a) What is the total mass of flour they have together?

(b) Do they have enough flour for the recipe? If not, how many kilograms of flour are they short of?

Working & Verification:

3. [Fraction of a Remaining Quantity]

A water tank has a capacity of 480 liters. On Monday, $\frac{3}{8}$ of the water was used. On Tuesday, $\frac{2}{5}$ of the **remaining** water was used. How many liters of water are left in the tank?

Step-by-step Solution:

Complete Answer Key & Pedagogical Notes

Section 1 (Equivalent):

- (a) 12 (3×4); (b) 56 (8×7); (c) 3 ($18 \div 6$);
(d) $\frac{3}{4}$ (divide by 15); (e) $\frac{2}{3}$ (divide by 14); (f) 16 (4×4).

Section 3 (Add/Subtract):

- (a) $\frac{15}{20} + \frac{8}{20} = \frac{23}{20} = 1\frac{3}{20}$;
(b) $\frac{20}{24} - \frac{9}{24} = \frac{11}{24}$;
(c) $2\frac{5}{15} + 1\frac{9}{15} = 3\frac{14}{15}$;
(d) $2\frac{3}{4} - 1\frac{3}{4} = 1\frac{2}{4} = 1\frac{1}{2}$ (borrow 1 as $\frac{4}{4}$);
(e) $3\frac{6}{7} - 2\frac{5}{7} = 1\frac{1}{7}$;
(f) $\frac{14}{20} - \frac{5}{20} + \frac{8}{20} = \frac{17}{20}$.

Section 2 (Conversions):

- (a) $2\frac{4}{24} = 2\frac{1}{6}$ (Student Exam check: do not flip quotient & remainder!);
(b) $6\frac{2}{6} = 6\frac{1}{3}$; (c) $9\frac{2}{5}$;
(d) $\frac{31}{7}$ ($4 \times 7 + 3$); (e) $\frac{47}{9}$; (f) $6\frac{3}{12} = 6\frac{1}{4}$.

Section 4 & 5 (Mult/Div & Applications):

- Sec 4: (a) $\frac{1}{6}$; (b) $\frac{20}{3} = 6\frac{2}{3}$; (c) $\frac{7}{10} \times \frac{15}{14} = \frac{3}{4}$; (d) $6 \times \frac{4}{3} = 8$;
(e) $\frac{9}{4} \times \frac{4}{3} \times \frac{10}{9} = \frac{10}{3} = 3\frac{1}{3}$.
Sec 5: 1. (a) $\frac{5}{12} = \frac{20}{48} < \frac{21}{48} = \frac{7}{16}$; (b) $1 - \frac{1}{12} < 1 - \frac{1}{13} \implies \frac{11}{12} < \frac{12}{13}$; (c) $\frac{7}{9} > \frac{7}{11}$.
2. (a) $\frac{3}{5} + 1\frac{1}{2} = \frac{6}{10} + 1\frac{5}{10} = 2\frac{1}{10}$ kg; (b) No; $2\frac{1}{4} - 2\frac{1}{10} = 2\frac{5}{20} - 2\frac{2}{20} = \frac{3}{20}$ kg short.
3. Remaining after Mon: $480 \times (1 - \frac{3}{8}) = 300$ L.
Tue used: $300 \times \frac{2}{5} = 120$ L. Left: $300 - 120 = 180$ L.